decreased winter snowfall, and a lengthening of the summer dry period. Since the latter half of the 20th century, otherwise healthy forests have experienced a steady increase in the percentage of trees that die each year. Higher temperatures and more frequent droughts also increase the likelihood of fires (see Figure 55.8). In boreal forests of western North America and Russia, for example, fires have burned twice the usual area in recent decades, again leading to widespread tree mortality. As the climate continues to warm, other changes in the geographic distribution of precipitation are likely, such as agricultural areas of the central United States becoming much drier.

Climate change has already affected many other ecosystems as well. In Europe and Asia, for example, plants are producing leaves earlier in the spring, while in tropical regions, the growth and survival of some species of coral have declined as water temperatures have risen. Recently, Australia's Great Barrier Reef had a catastrophic decline: The combined effects of two consecutive marine heat waves (one in 2016, another in 2017) killed so many corals that two-thirds of the reef was severely degraded—so degraded that some scientists worry it may never recover. A key take-home message from these examples is that a given effect of climate change may, in turn, cause a series of other biological changes. The exact nature of such cascading biological effects can be hard to predict, but it is clear that the more our planet warms, the more severely its ecosystems will be affected.

Modeling Climate Change

At the current rates that human activities are adding CO_2 and other greenhouse gases to the atmosphere, global models predict that Earth's temperature will rise by an additional 3°C (5°F) by the end of the 21st century. What data are used to construct such models, and are their predictions accurate?

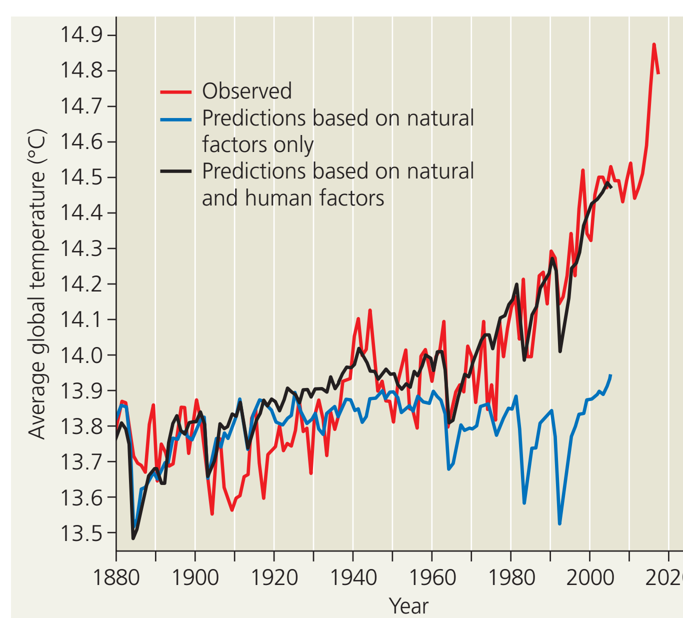
Global models are constructed using data on factors that affect the absorption of solar radiation at Earth's surface. These data are crucial because global temperatures rise when more solar radiation is absorbed by Earth's surface, and they fall when less solar radiation is absorbed by the surface. Natural factors that affect the absorption of solar radiation include an 11-year solar cycle and volcanic explosions. During each round of the solar cycle, the amount of energy emitted by the sun increases and decreases in a regular way, thereby contributing to a well-understood rise or fall in global temperatures at different points in time. Volcanic explosions emit gases and tiny particles that block incoming solar radiation, thereby contributing to a temporary drop in global temperatures. (Changes in the shape of Earth's orbit around the sun also contribute to increases or decreases in global temperatures, but those changes occur gradually over tens of thousands of years and hence are not a cause of recent climate change.)

Scientists also have collected vast amounts of data on how human activities affect absorption of solar radiation. When we burn fossil fuels for electricity, heat, and transportation, we cause CO_2 and other greenhouse gases to be released to the atmosphere. As shown in Figure 56.30, adding CO_2 and other greenhouse gases to the atmosphere increases the absorption of solar radiation, leading to global warming. Other human activities decrease the absorption of solar radiation, thereby tending to reduce global temperatures. For example, dust released by plowing fields and the emission of gases such as sulfur dioxide (SO_2) both decrease the absorption of solar energy, thus reducing global temperatures.

Each of these and other types of data provides a single, well-understood piece of a grand puzzle—how Earth's temperature is affected by a combination of natural and human factors. Scientists use computer models to put the pieces of this puzzle together. Each year, large amounts of data on natural and human factors that affect the absorption of solar radiation are entered into the model, which then sums their effects to predict Earth's temperature. Over time, predictions from these models have become increasingly accurate: Global climate models can now reproduce observed changes such as the increase in global warming that has occurred since the 1950s **(Figure 56.32)** and the global cooling that results from volcanic eruptions.

But why go to all this trouble developing models when Earth's temperature can be measured each year? The reason is that we have only one Earth and hence cannot perform

▼ Figure 56.32 Factors contributing to rising global temperatures. Scientists have measured the temperature of our planet since 1880. These observed temperatures (red) can be compared to results from climate models that incorporate only natural factors that affect global temperatures (blue) and from models that incorporate both natural and human factors (black).



VISUAL SKILLS *Describe how well the results shown in the blue and black curves match observed temperature changes over time. Use those results to evaluate whether human activities such as burning fossil fuels have contributed to observed temperature changes over time.*